

A LINEAR-TIME ALGORITHM FOR FINDING A COMPLETE GRAPH MINOR IN A DENSE GRAPH*

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Abstract. Let $g(t)$ be the minimum number such that every graph G with average degree $d(G) \geq g(t)$ contains a K_t -minor. Such a function is known to exist, as originally shown by Mader. Kostochka and Thomason independently proved that $g(t) \in \Theta(t\sqrt{\log t})$. This paper shows that for all fixed $\epsilon > 0$ and fixed sufficiently large $t \geq t(\epsilon)$, if $d(G) \geq (2 + \epsilon)g(t)$, then we can find this K_t -minor in linear time. This improves a previous result by Reed and Wood who gave a linear-time algorithm when $d(G) \geq 2^{t-2}$.

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1. Introduction. A major result in the theory of graph minors is that every graph G with sufficiently large average degree $d(G)$ contains a complete graph K_t as a minor. That is, a K_t can be constructed from G using *vertex deletion*, *edge deletion*, and *edge contraction*. Let

$$g(t) := \min\{D : \text{every graph } G \text{ with } d(G) \geq D \text{ contains a } K_t\text{-minor}\}.$$

Mader [4] showed that $g(t)$ is well-defined, and that $g(t) \leq 2^{t-2}$. Subsequently, Mader [5] improved this bound to $g(t) \leq 16t \log_2 t$, and later this was improved to $g(t) \in \Theta(t\sqrt{\log t})$ by Thomason [9] and Kostochka [2, 3], which is best possible. Thomason [10] later determined the asymptotic constant for this bound.

This paper considers linear-time algorithms for finding a K_t -minor in a graph with high average degree. This question was first considered by Reed and Wood [6] who gave a $O(n)$ time algorithm to find a K_t -minor in an n -vertex graph G with $d(G) \geq 2^{t-2}$. We improve on this result by lowering the required bound on the average degree to within a constant factor of optimal.

THEOREM 1.1. *For all fixed $\epsilon > 0$ and fixed sufficiently large $t \geq t(\epsilon)$, there is a $O(n)$ time algorithm that, given an n -vertex graph G with average degree $d(G) \geq (2 + \epsilon)g(t)$, finds a K_t -minor in G .*

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Reed and Wood used their algorithm mentioned above as a subroutine for finding separators in H -minor free graphs (also see [12] for a related separator result). This result has subsequently been used in other algorithms for H -minor free graphs, in particular, shortest path algorithms by Tazari and Müller-Hannemann [8] and Wulff-Nilsen [11], and a maximum matching algorithm by Yuster and Zwick [13]. The algorithm presented in this paper speeds up all these results (in terms of the dependence on H).

Finally, note that Robertson and Seymour [7] describe a $O(n^3)$ time algorithm that tests whether a given n -vertex graph contains a fixed graph H as a minor. The time complexity was improved to $O(n^2)$ by Kawarabayashi, Kobayashi, and Reed [1]. Kawarabayashi and Reed have announced a $O(n \log n)$ time algorithm for this problem.

2. Algorithm. Given a vertex v of a graph G , we denote by $\deg_G(v)$ and $N_G(v)$ the degree and neighborhood of v in G , respectively. We drop the subscript when G is clear from the context. Define a *matching* $M \subseteq E(G)$ to be a set of edges such that no two edges in M share an endpoint. Let $V(M)$ be the set of endpoints of the edges in M . An *induced matching* in G is a matching such that any two vertices x, y of $V(M)$ are only adjacent in G when $xy \in M$. Given a matching M in G , let G/M be the graph formed by contracting each edge of M in G .

We fix $\epsilon > 0$ and $t \geq 3$ such that $g(t) \geq \max\{t, \frac{2t}{\epsilon}\}$. We may assume $t \geq 3$ since finding a K_1 - or K_2 -minor is trivial, and that $g(t) \geq \max\{t, \frac{2t}{\epsilon}\}$ for sufficiently large t , since $g(t) \in \Theta(t\sqrt{\log t})$. Consider the following algorithm that takes as input a graph given as a list of vertices and a list of edges. The implicit output of the algorithm is the sequence of contractions and deletions that produce a K_t -minor.

3. Correctness of algorithm. First, we prove that $\text{FINDMINOR}(G)$ does output a K_t -minor. Define $m := |E(G)|$. We must ensure the following: that FINDMINOR finds a K_t -minor in steps 5 and 8; that $S' - A \neq \emptyset$ in step 9; that the graph constructed in step 10 contains a K_t subgraph; and that our exhaustive search in step 11 finds a K_t -minor of G .

Consider step 5. Assume that FINDMINOR finds a K_t -minor in any graph G' with $|V(G')| < n$ where $d(G') \geq (2 + \epsilon)g(t)$. Consider the induced matching M' . Contracting any single edge vw of M' does not lower the average degree, as we only lose $|N(v) \cap N(w)| + 1 \leq \frac{1}{2}d(G)$ edges and one vertex. Since the matching is induced, contracting every edge in M' does not lower the average degree. Since $|M| > \frac{1}{8d(G)}n$, M is not empty and M' is not empty. Thus $d(G') \geq d(G) \geq (2 + \epsilon)g(t)$ and $|V(G')| < |V(G)| = n$. Thus, by induction, running the algorithm on G' finds a K_t -minor, and as such we find one for G .

If we recurse at step 8, then $2|A| \geq d(G)|B'|$ and $B' \neq \emptyset$. Now $|V(G'[B'])| = |B'|$ and $|E(G'[B'])| \geq |A|$, since every assigned vertex corresponds to an edge of $G'[B']$. Thus

$$d(G'[B']) = \frac{2|E(G'[B'])|}{|V(G'[B'])|} \geq \frac{2|A|}{|B'|} \geq d(G).$$

Also, $|V(G'[B'])| = |B'| < n$, since otherwise $A = S' = \emptyset$, contradicting $2|A| \geq d(G)|B'| > 0$. Hence, by assumption, the algorithm will find a K_t -minor in $G'[B']$. Thus the algorithm finds a K_t -minor for G .

Now we show that $|S'| > |A|$ in step 9. We have $2|A| < d(G)|B'|$ or $B' = \emptyset$. First consider the case when $2|A| < d(G)|B'|$. Note that $2m = d(G)n$, and that

Algorithm 1. FINDMINOR (input: n -vertex graph G with $d(G) \geq (2 + \epsilon)g(t)$)

- 1: Delete edges of G so that $(2 + \epsilon)g(t) \leq d(G) \leq (2 + \epsilon)g(t) + 1$.
 - 2: Delete vertices of low degree so that the minimum degree $\delta(G) > \frac{1}{2}d(G)$.
 - 3: Let $S := \{v \in V(G) : \deg(v) \leq d(G)^2\}$, and let $B := \{v \in V(G) : \deg(v) > d(G)^2\}$.
 [Note that B is possibly empty, and that S and B partition $V(G)$.]
 - 4: Say an edge $vw \in E(G)$ is *good* if $v, w \in S$ and $|N(v) \cap N(w)| \leq \frac{1}{2}(d(G) - 2)$.
 Greedily construct a maximal matching M of good edges.
 [Note that it is possible that no edges are good, in which case $M = \emptyset$.]
 - 5: If $|M| > \frac{1}{8d(G)}n$, then greedily construct a maximal induced submatching M' of M . That is, initialize $M' := \emptyset$ and $Q := M$, and repeat the following algorithm until $Q = \emptyset$: pick an edge $vw \in Q$, add vw to M' , and delete from Q the edge vw and every edge with an endpoint adjacent to v or w .
 Let $G' := G/M'$. Run FINDMINOR(G') and stop.
 - 6: Now assume $|M| \leq \frac{1}{8d(G)}n$. Let $B' := B \cup V(M)$ and $S' := S - V(M)$.
 [Note that, similarly to step 3, S' and B' partition $V(G)$.]
 - 7: Greedily compute a maximal subset A of S' such that each vertex $u \in A$ is *assigned* to a pair of vertices in $N(u) \cap B'$, and each pair of vertices in B' has at most one vertex in A assigned to it.
 - 8: If $2|A| \geq d(G)|B'|$ and $B' \neq \emptyset$, then let G' be the graph obtained from G as follows: For each pair of distinct vertices $x, y \in B'$ with an assigned vertex $z \in A$, contract the edge xz .
 Run FINDMINOR($G'[B']$) and stop.
 - 9: Now assume $2|A| < d(G)|B'|$ or $B' = \emptyset$. Choose $v \in S' - A$.
 [We prove below that $S' - A \neq \emptyset$. Since v is not assigned, for every pair x, y of vertices in $N(v) \cap B$ some vertex $z \in A$ is assigned to x, y .]
 - 10: If $|N(v) \cap B'| \geq t$, then let G' be the graph obtained from G as follows: For each pair of distinct vertices $x, y \in N(v) \cap B'$, if z is the vertex in A assigned to x and y , then contract xz into x (so that the new vertex is in B'). Then $G'[N(v) \cap B'] \supseteq K_t$. Stop.
 - 11: Otherwise let $G' := G[\{v\} \cup (N_G(v) \cap S')]$ and run an exhaustive search to find a K_t -minor in G' .
 [Below we prove that $d(G) \geq g(t)$ and $|V(G')| \leq d(G)^2 + 1$.]
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$d(G)^2|B| < \sum_{v \in B} \deg(v) \leq 2m$, and so $|B| < \frac{2m}{d(G)^2} = \frac{1}{d(G)}n$. Now $|S'| = |S| - 2|M| \geq |S| - \frac{1}{4d(G)}n$ by step 6. Thus,

$$|S'| \geq |S| - \frac{1}{4d(G)}n = (n - |B|) - \frac{1}{4d(G)}n > n - \frac{1}{d(G)}n - \frac{1}{4d(G)}n = \frac{4d(G) - 5}{4d(G)}n.$$

By steps 9 and 6,

$$|A| < \frac{d(G)}{2}|B'| = \frac{d(G)}{2}(|B| + 2|M|) < \frac{d(G)}{2} \left(\frac{1}{d(G)}n + \frac{1}{4d(G)}n \right) = \frac{5}{8}n.$$

Thus, if $|S'| \leq |A|$, then $\frac{4d(G)-5}{4d(G)}n < \frac{5}{8}n$, so $3d(G) < 10$ and $d(G) \leq 3$. This is a contradiction since $d(G) \geq (2 + \epsilon)g(t) > 2g(3) = 4$. (We have $g(t) \geq g(3) = 2$, since $g(t)$ is nondecreasing.) Hence, $|S'| > |A|$. Now consider the case that $B' = \emptyset$. Then

$|S'| = n$ and $A = \emptyset$, since the vertices of A are assigned to pairs of vertices in B' . Hence $|S'| > |A|$.

Now consider step 10. $G'[N(v) \cap B']$ has at least t vertices by assumption. Each pair of distinct vertices x, y in $N(v) \cap B'$ has an assigned vertex in A , as otherwise v would have been assigned to x and y . Hence the vertex z exists, and x and y are adjacent after contracting xz . Therefore, all pairs of vertices in $N(v) \cap B'$ become adjacent, and $G'[N(v) \cap B']$ is a complete graph, and we have found our K_t -minor in G .

Finally consider step 11. G' is an induced subgraph of G , and so if we can find K_t as a minor in G' , we have a K_t -minor in G . We use an exhaustive search, so all we need to ensure is that G' does have a K_t -minor. Thus, we simply need to ensure that $d(G') \geq g(t)$. By steps 1 and 2, $\deg_G(v) > \frac{1}{2}d(G) \geq \frac{\epsilon}{2}g(t) \geq t$. Since step 10 was not applicable, v has at most $t-1$ neighbors in B' . Thus v has some neighbor in S' . Let w be a vertex of $G' - v$. Thus vw is an edge and $v, w \in S'$. Since neither v nor w was matched by M , and since M is maximal, vw is not good. Since $v, w \in S' \subseteq S$, this means that $|N(v) \cap N(w)| > \frac{1}{2}(d(G) - 2)$. As v has at most $t-1$ neighbors in B' , we have $|N(v) \cap N(w) \cap S'| > \frac{1}{2}(d(G) - 2) - (t-1)$. Every common neighbor of v and w in S' is a neighbor of w in G' , by definition, so $\deg_{G'}(w) > \frac{1}{2}(d(G) - 2) - (t-1)$. Since v is dominant in G' , we have $d(G') \geq \frac{1}{2}(d(G) - 2) - (t-1)$, which is at least $g(t)$ as required since $d(G) \geq (2 + \epsilon)g(t)$ and $\epsilon g(t) \geq 2t$.

4. Time complexity. Now that we have shown that FINDMINOR will output a K_t -minor, we must ensure it does so in $O(n)$ time (for fixed t and ϵ).

First, suppose FINDMINOR runs without recursing. Recall that our input graph G is given as a list of vertices and a list of edges, from which we will construct adjacency lists as it is read in. Since our goal in step 1 is to ensure that $m \leq \frac{1}{2}((2 + \epsilon)g(t) + 1)n$, we can do this by taking, at most, the first $\frac{1}{2}((2 + \epsilon)g(t) + 1)n$ edges, and ignoring the rest. This can be done in $O(n)$ time, and from now on we may assume that $m \in O(n)$. In step 2, since we are only deleting vertices of bounded degree, this can be done in $O(n)$ time. Clearly, steps 3, 6, and 9 can be implemented in $O(n)$ time. By definition, the degree of any vertex in S or S' is at most $((2 + \epsilon)g(t) + 1)^2$. Hence steps 4, 5, 7, 8, and 10 take $O(n)$ time. Finally, for step 11 note that $|V(G')| \leq d(G)^2 + 1$, so exhaustive search runs in $O(1)$ time for fixed t . Hence the algorithm without recursion runs in $O(n)$ time.

Should FINDMINOR recurse, we need to ensure that the order of the graph we recurse on is a constant factor less than n . Then the overall time complexity is $O(n)$ (by considering the sum of a geometric series). In step 5, the endpoints of edges in M have degree less than or equal to $d(G)^2$, and so $|M'| \geq \frac{1}{2d(G)^2}|M| \geq \frac{1}{16d(G)^3}n$. This ensures that $|V(G')| \leq (1 - \frac{1}{16d(G)^3})n$, as desired. In step 8, the order of $G'[B']$ is at most $\frac{2|A|}{d(G)} \leq \frac{2n}{d(G)}$. Hence it follows that the overall time complexity is $O(n)$.

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